

Software Requirements Specification (SRS)

Library Management System (LMS)

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the **Library Management System (LMS)**. This specification is intended for developers, project managers, and evaluators.

1.2 Scope

The LMS is a web-based application designed to automate primary library operations. The system allows users to search for books, reserve books, and manage their profile, while administrators can manage book inventories, track issued books, and oversee user accounts.

1.3 Definitions & Acronyms

- **SRS:** Software Requirements Specification
- **LMS:** Library Management System
- **Admin:** System Administrator with elevated privileges
- **Member:** Registered library user/student

2. Overall Description

2.1 Product Perspective

The LMS is an independent, web-based system operating on a client-server architecture with a relational database backend (MySQL).

2.2 User Classes and Characteristics

1. **Member / Student:** Can search the catalog, view book availability, borrow/reserve books, and view account history.
2. **Administrator / Librarian:** Can add/edit/delete books, approve borrow requests, manage user profiles, and view system reports.

2.3 Operating Environment

- **Server:** Apache / Nginx web server, PHP 8.x, MySQL 8.0 database
- **Client:** Any standard web browser (Google Chrome, Mozilla Firefox, Edge)

3. System Features & Functional Requirements

3.1 User Authentication

- **FR-1.1:** The system shall allow users to register with a name, email, and password.
- **FR-1.2:** The system shall allow registered users to log in securely using their credentials.
- **FR-1.3:** The system shall support role-based access control (Admin vs. Member).

3.2 Catalog & Search Management

- **FR-2.1:** The system shall allow all users to search for books by title, author, or category.
- **FR-2.2:** The system shall display real-time availability status for each book (Available / Issued).

3.3 Book Management (Admin Only)

- **FR-3.1:** The admin shall be able to add new books with details (Title, Author, ISBN, Category, Quantity).
- **FR-3.2:** The admin shall be able to update or remove existing book records.

3.4 Borrowing & Returns

- **FR-4.1:** A logged-in member shall be able to request a book checkout.
- **FR-4.2:** The system shall update book inventory automatically upon successful checkout or return.
- **FR-4.3:** The system shall record issue date and expected return date.

4. Non-Functional Requirements

4.1 Performance Requirements

- The system shall load pages within 2 seconds under normal network conditions.
- The system shall support up to 100 concurrent active users without degradation of service.

4.2 Security Requirements

- User passwords must be hashed using `bcrypt` before storage.
- All admin actions must require authenticated session validation.
- Input fields must be sanitized to prevent SQL Injection and Cross-Site Scripting (XSS).

4.3 Availability & Reliability

- The system shall maintain 99% uptime during operational library hours.
- Data backups must be performed daily.

4.4 Usability

- The user interface must be responsive across desktop and tablet screen sizes.
- Navigation should require no more than 3 clicks to reach any primary function.

5. System Diagrams (Context & Workflow)

5.1 Use Case

